

# Operations on graphs



# Outline

Operations on graphs

Operations related to  $y_1 = f(x) + a$

Operations related to  $y_2 = f(x + a)$

Operations related to  $y_3 = k f(x)$

Operations related to  $y_4 = f(k x)$

Operations related to  $y_5 = |f(x)|$

Operations related to  $y_6 = f(|x|)$

# Operations on graphs

# Introduction: Operations on graphs

- ▶ Knowing the graph of a function  $y = f(x)$ , we want to draw, without computations, the graphs of functions obtained from  $f$  through simple operations
- ▶ These operations correspond to **translations**, **dilations** and **reflections** of the graph
- ▶ It is a fundamental tool to quickly draw graphs starting from those of the elementary functions

# The six operations

Knowing the graph of a function  $y = f(x)$ , by means of simple geometric transformations it is possible to draw the graph of the following functions:

$$y_1 = f(x) + a, \quad a \in \mathbb{R}$$

$$y_2 = f(x + a), \quad a \in \mathbb{R}$$

$$y_3 = k f(x), \quad k \in \mathbb{R}$$

$$y_4 = f(k x), \quad k \in \mathbb{R}$$

$$y_5 = |f(x)|$$

$$y_6 = f(|x|)$$

# Translations, dilations and reflections

- ▶ Therefore, starting from the elementary functions, it is possible to build, by means of these operations, a large variety of new functions.
- ▶ These operations create **translations**, **dilations** and **reflections** of the original function.

# Summary

- ▶ From the graph of  $y = f(x)$  we obtain the graphs of  $f(x) + a$ ,  $f(x + a)$ ,  $k f(x)$ ,  $f(kx)$ ,  $|f(x)|$  and  $f(|x|)$
- ▶ The operations produce translations, dilations and reflections
- ▶ In the next sections we study them one at a time

# Operations related to

$$y_1 = f(x) + a$$



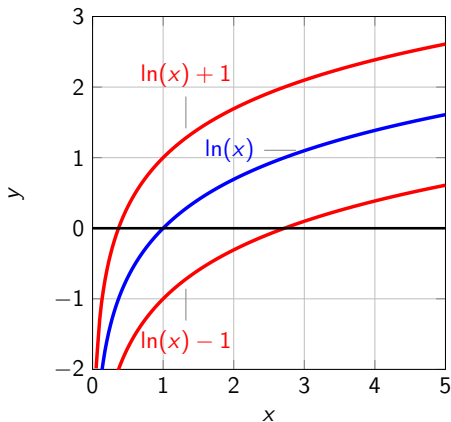
## Introduction: Operations related to $y_1 = f(x) + a$

- ▶ We see what happens to the graph when we **add a constant  $a$**  to the values of the function
- ▶ All the values  $f(x)$  are increased (or decreased) by the same amount: we expect a **vertical** shift of the graph

## Example

Consider  $y = \ln x$ . Then

$$y_1 = \ln(x) + a \quad (x > 0).$$



# Vertical translation

- ▶ The graph of  $y_1$  is obtained from that of  $y$  by a **translation** of  $a$  units **upward** if  $a > 0$ , **downward** if  $a < 0$ .

# Summary

$$y_1 = f(x) + a:$$

- ▶ Vertical translation by  $a$  units
- ▶ Upward if  $a > 0$ , downward if  $a < 0$
- ▶ The domain does not change (e.g.  $\ln(x) + a$  is defined for  $x > 0$ )

# Operations related to

$$y_2 = f(x + a)$$

## Introduction: Operations related to $y_2 = f(x + a)$

- ▶ We see what happens to the graph when we **add a constant**  $a$  to the variable  $x$ , before applying  $f$
- ▶ We expect a **horizontal** shift of the graph, which also changes the domain of the function

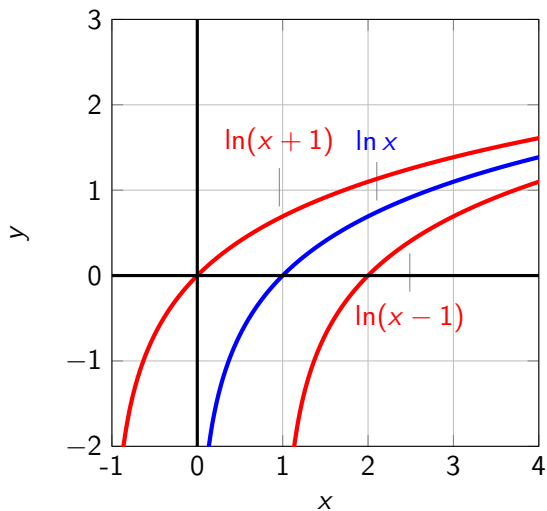
## Example (1/2)

Consider again  $y = \ln x$ . Then

$$y_2 = \ln(x + a).$$

- ▶ Since  $\ln x$  is defined for  $x > 0$ ,  $\ln(x + a)$  will be defined for  $x + a > 0$ , i.e., for  $x > -a$ .
- ▶ Since  $\ln x = 0$  if  $x = 1$ , we have  $\ln(x + a) = 0$  for  $x + a = 1$ , i.e.,  $x = 1 - a$ .

## Example (2/2)



# Horizontal translation

- ▶ The graph of  $y_2$  is obtained from that of  $y$  by a **translation** of  $a$  units to the **left** if  $a > 0$ , to the **right** if  $a < 0$ .

# Summary

$$y_2 = f(x + a):$$

- ▶ Horizontal translation by  $a$  units
- ▶ To the left if  $a > 0$ , to the right if  $a < 0$
- ▶ The domain shifts:  $\ln(x + a)$  is defined for  $x > -a$  and vanishes at  $x = 1 - a$

# Operations related to

$$y_3 = k f(x)$$

## Introduction: Operations related to $y_3 = k f(x)$

- ▶ We see what happens to the graph when we **multiply** the values of the function by a constant  $k$
- ▶ We expect the graph to be “stretched” or “squeezed” in the **vertical** direction, and possibly reflected

## Vertical dilation and contraction (1/2)

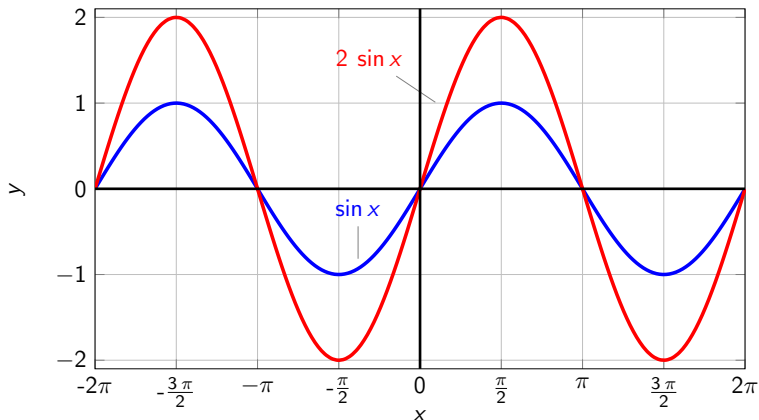
- ▶ The graph of  $y_3 = k f(x)$  is obtained from that of  $f$  by multiplying all the ordinates  $f(x)$  by  $k$ .
- ▶ In particular, if  $k = -1$  the ordinates simply change sign, so that the graph of  $y_3$  is symmetric to that of  $f$  with respect to the  $x$ -axis.
- ▶ Note that if  $k > 1$ , the graph is “**stretched**” in the **vertical direction**, dilating the **positive** ordinates **upward** and the **negative** ones **downward**.

## Vertical dilation and contraction (2/2)

- ▶ Conversely, if  $0 < k < 1$  the graph “**shrinks**”, again in the vertical direction.
- ▶ Therefore, the operation of multiplying  $f(x)$  by  $k$  has the geometric meaning of a **dilation** (if  $|k| > 1$ ) or a **contraction** (if  $|k| < 1$ ) along the  $y$ -axis, possibly accompanied by a **reflection with respect to the  $x$ -axis**, if  $k < 0$ .

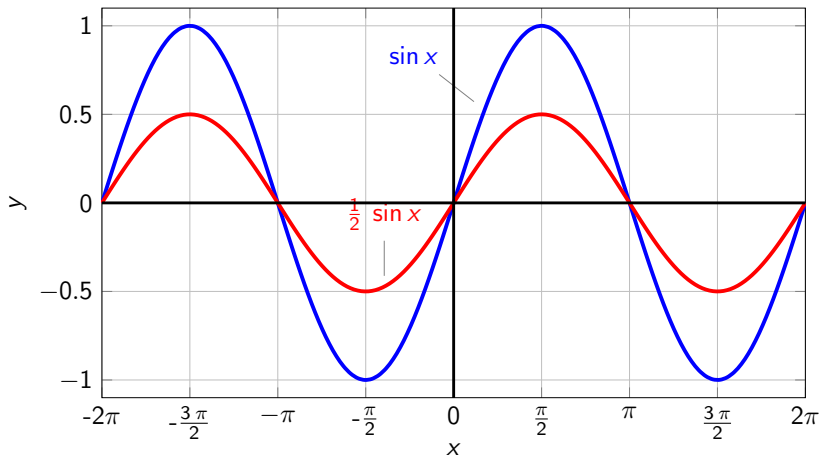
## Example (1/3)

Consider  $y = \sin x$ , then  $y_3 = k \sin(x)$ . With  $k = 2$ :



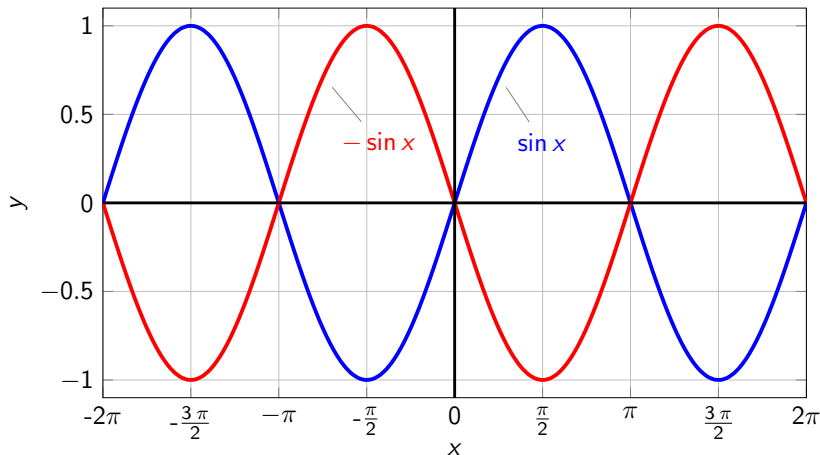
## Example (2/3)

With  $k = \frac{1}{2}$ :



## Example (3/3)

With  $k = -1$ :



# Summary

$$y_3 = k f(x):$$

- ▶ All the ordinates are multiplied by  $k$
- ▶  $|k| > 1$ : vertical dilation;  $|k| < 1$ : vertical contraction
- ▶  $k < 0$ : also a reflection with respect to the  $x$ -axis (for  $k = -1$  only a reflection)

# Operations related to

$$y_4 = f(kx)$$

## Introduction: Operations related to $y_4 = f(kx)$

- ▶ We see what happens to the graph when we **multiply the variable**  $x$  by a constant  $k$ , before applying  $f$
- ▶ We expect a **horizontal change of scale**: the graph is compressed or dilated along the  $x$ -axis, and possibly reflected

## Horizontal change of scale ( $1/2$ )

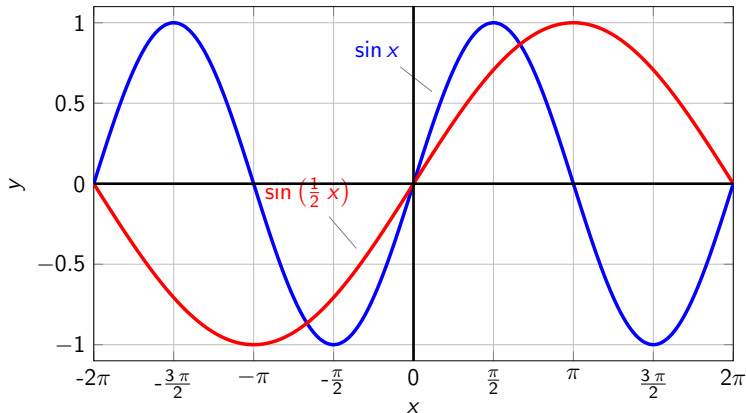
- ▶ The graph of  $y_4 = f(kx)$  is obtained from that of  $f(x)$  by a **change of scale** along the  $x$ -axis.
- ▶ If  $k > 1$ ,  $kx$  grows faster than  $x$ , and therefore the graph of  $y_4$  will be similar to that of  $f$  but with faster oscillations, that is, it will be “**compressed**” in the **horizontal direction** by a factor  $\frac{1}{k}$ .

## Horizontal change of scale (2/2)

- ▶ Similarly, if  $0 < k < 1$  the graph will appear “**dilated**” in the **horizontal direction**, with gentler oscillations.
- ▶ If  $k < 0$ , in addition to a compression (if  $|k| > 1$ ) or a dilation (if  $|k| < 1$ ) along the  $x$ -axis, there will be a **reflection with respect to the  $y$ -axis**.

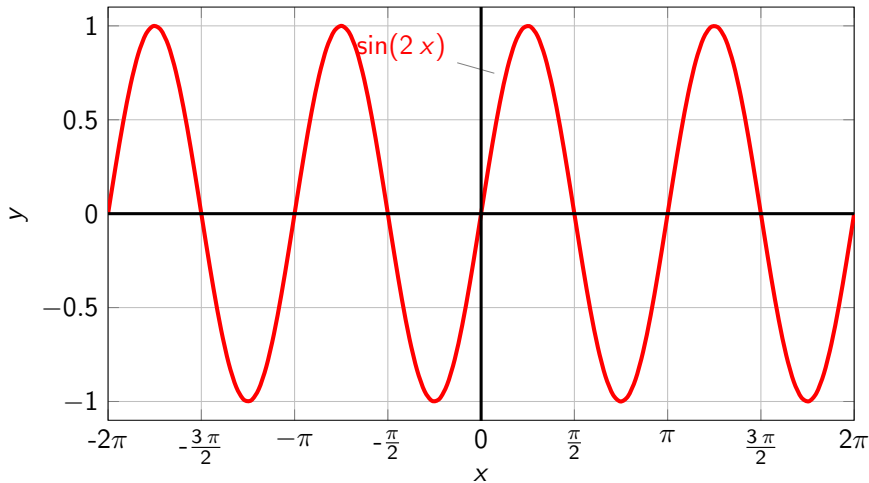
## Example (1/2)

Consider  $y = \sin x$ , then  $y_4 = \sin(kx)$ . With  $k = \frac{1}{2}$  and  $k = 2$  we have:



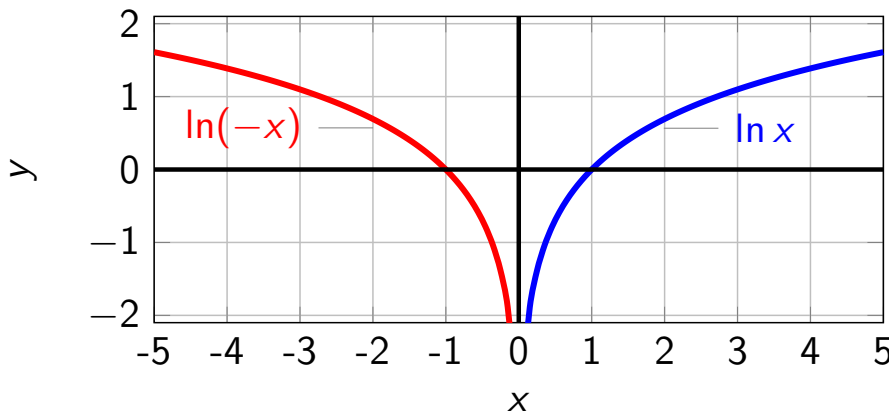
## Example (2/2)

With  $k = 2$ :



## Example

Consider  $y = \ln x$ , then  $y_4 = \ln(kx)$ . With  $k = -1$  we have:



# Summary

$$y_4 = f(kx):$$

- ▶ Change of scale along the  $x$ -axis
- ▶  $|k| > 1$ : horizontal compression by a factor  $\frac{1}{|k|}$ ;  $|k| < 1$ : horizontal dilation
- ▶  $k < 0$ : also a reflection with respect to the  $y$ -axis (e.g.  $\ln(-x)$ )

# Operations related to

$$y_5 = |f(x)|$$

## Introduction: Operations related to $y_5 = |f(x)|$

- ▶ We see how to obtain the graph of the **absolute value** of a function
- ▶ The absolute value makes all values non-negative: the part of the graph below the  $x$ -axis is “flipped” upward

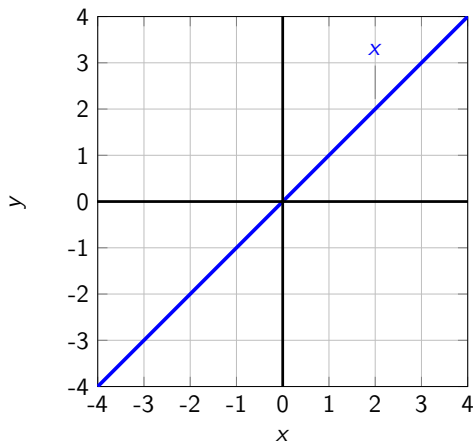
## Absolute value of $f(x)$

$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0, \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

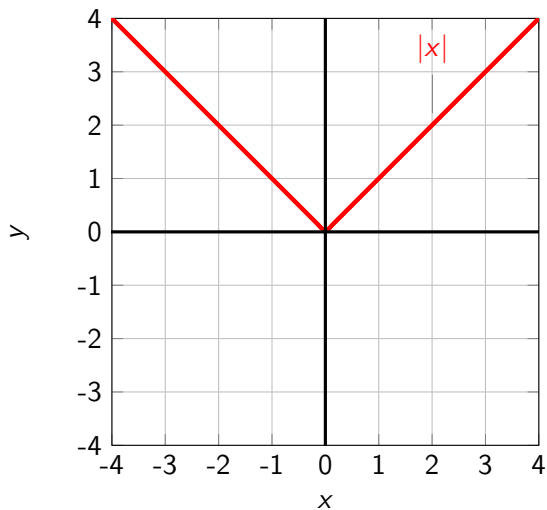
- ▶ When passing from the graph of  $y = f(x)$  to that of  $y_5 = |f(x)|$ , the points with **non-negative ordinate remain unchanged**, while those with **negative ordinate are transformed into their symmetric points with respect to the  $x$ -axis**.
- ▶ The graph of  $y_5 = |f(x)|$  is obtained from that of  $f$  by “flipping” symmetrically with respect to the  $x$ -axis the part of the graph of  $f$  that lies in the lower half-plane, and leaving the rest unchanged.

## Example (1/2)

Consider  $y = x$ , then  $y_5 = |x|$

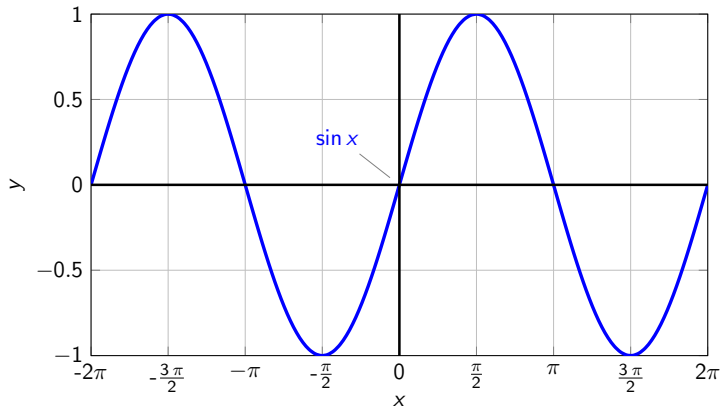


## Example (2/2)

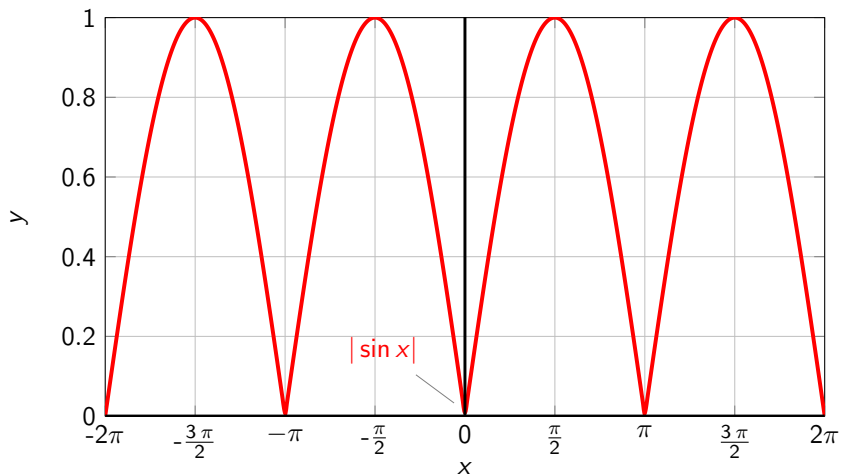


## Example (1/2)

Consider  $y = \sin x$ , then  $y_5 = |\sin x|$



## Example (2/2)



# Summary

$$y_5 = |f(x)|:$$

- ▶ The points with non-negative ordinate remain unchanged
- ▶ The part of the graph in the lower half-plane is flipped with respect to the  $x$ -axis
- ▶ The graph of  $|f(x)|$  lies entirely in the half-plane  $y \geq 0$

# Operations related to

$$y_6 = f(|x|)$$

## Introduction: Operations related to $y_6 = f(|x|)$

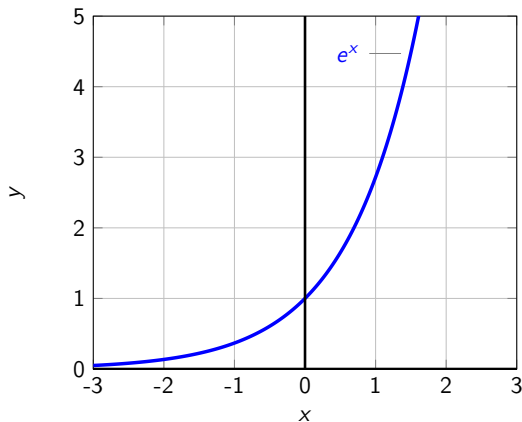
- ▶ Finally, we see the case in which the absolute value is applied to the **variable**  $x$
- ▶ We always obtain an **even** function, whose graph is symmetric with respect to the  $y$ -axis

## Graph of $y_6 = f(|x|)$

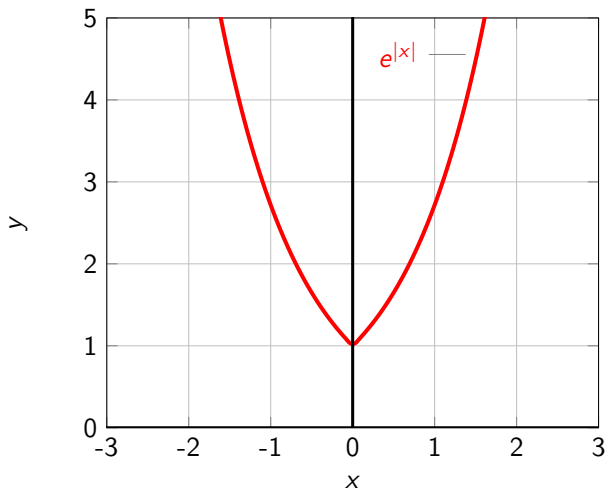
- ▶ Finally, to draw the graph of  $y_6 = f(|x|)$ , we observe that  $|x| = x$  for  $x \geq 0$ , and hence in the right half-plane the two graphs coincide.
- ▶ We have  $|-x| = |x|$ , and hence  $y_6$  is an even function, therefore symmetric with respect to the  $y$ -axis.
- ▶ Consequently, the graph of  $y_6 = f(|x|)$  is drawn by **leaving the graph of  $f$  unchanged in the right half-plane and flipping it symmetrically with respect to the  $y$ -axis.**

## Example (1/2)

Consider  $y = e^x$ , then  $y_6 = e^{|x|}$



## Example (2/2)



# Summary

$$y_6 = f(|x|):$$

- ▶ In the right half-plane ( $x \geq 0$ ) the graph coincides with that of  $f$
- ▶  $f(|x|)$  is even: the right part of the graph is flipped with respect to the  $y$ -axis
- ▶ Example:  $e^{|x|}$

# Summary (1/2)

## Translations:

- ▶  $f(x) + a$ : vertical translation (upward if  $a > 0$ , downward if  $a < 0$ )
- ▶  $f(x + a)$ : horizontal translation (to the left if  $a > 0$ , to the right if  $a < 0$ )

## Dilations and reflections:

- ▶  $k f(x)$ : vertical dilation/contraction, reflection with respect to the  $x$ -axis if  $k < 0$
- ▶  $f(kx)$ : horizontal compression/dilation, reflection with respect to the  $y$ -axis if  $k < 0$

## Summary (2/2)

### Absolute values:

- ▶  $|f(x)|$ : the part of the graph below the  $x$ -axis is flipped with respect to the  $x$ -axis
- ▶  $f(|x|)$ : the part for  $x \geq 0$  is kept and flipped with respect to the  $y$ -axis
- ▶ Combining these operations, we can draw the graphs of many functions starting from those of the elementary functions